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35. Elements a, b, c and d have the following electronic configurations:

    a: $1s^2$, $2s^2$, $2p^6$

    b: $1s^2$, $2s^2$, $2p^6$, $3s^2$, $3p^1$

    c: $1s^2$, $2s^2$, $2p^6$, $3s^2$, $3p^6$

    d: $1s^2$, $2s^2$, $2p^1$

    Which elements among these will belong to the same group of periodic table

36. Give the general electronic configuration of lanthanides and actinides?

37. Why halogens act as oxidising agents?

38. Mention any two anomalous properties of second period elements.

39. Explain the pauling method for the determination of ionic radius.

40. Explain the periodic trend of ionisation potential.

41. Explain the diagonal relationship.

42. Why the first ionisation enthalpy of sodium is lower that that of magnesium while its second ionisation enthalpy is higher than that of magnesium?

43. By using paulings method calculate the ionic radii of K^+ and Cl^- ions in the potassium chloride crystal. Given that $d_{K^+-Cl^-} = 3.14$ A

44. Explain the following, give appropriate reasons.

    (i) Ionisation potential of N is greater than that of O

    (ii) First ionisation potential of C-atom is greater than that of B atom, where as the reverse is true is for second ionisation potential.

    (iii) The electron affinity values of Be, Mg and noble gases are zero and those of N (0.02 eV) and P (0.80 eV) are very low

    (iv) The formation of F^- (g) from F(g) is exothermic while that of O^{2-}(g) from O (g) is endothermic.

45. What is screening effect? Briefly give the basis for pauling's scale of electronegativity.

46. State the trends in the variation of electronegativity in group and periods.